



Selection Process

Workshop Foundation

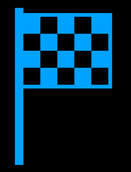
Apple Developer Academy | SENAC

adapted from https://www.hackingwithswift.com

Agenda







01 Introduction

02 Control

03 Collections

04 Functions

05 Challenge

Agenda



01 Introduction

02 Control

03 Collections

04 Functions

05 Challenge

Set

Arrays

Dictionary

Tuples

Default

Creating Arrays

```
var beatles = ["John", "Paul", "George", "Ringo"]
let numbers = [4, 8, 15, 16, 23, 42]
var temperatures = [25.3, 28.2, 26.4]
```

Accessing Array Values

```
print(beatles[0])
print(numbers[1])
print(temperatures[2])
```

⚠️ **Tip:** Make sure an item exists at the index you're asking for, otherwise your code will crash – your app will just stop working.

Add data in array

```
beatles.append("Allen")
beatles.append("Adrian")
beatles.append("Novall")
beatles.append("Vivian")
```

Creating typed arrays

```swift
var scores = Array<Int>()
scores.append(100)
scores.append(80)
scores.append(85)
print(scores[1])
```

OR

```swift
var albums = [String]()
albums.append("Folklore")
albums.append("Fearless")
albums.append("Red")
```

Array operations

```
var characters = ["Lana", "Pam", "Ray", "Sterling"]
print(characters.count)

characters.remove(at: 2)
print(characters.count)

characters.removeAll()
print(characters.count)
```

Array operations

```
let bondMovies = ["Casino Royale", "Spectre", "No Time To Die"]
print(bondMovies.contains("Frozen"))
```

Array operations

```
let cities = ["London", "Tokyo", "Rome", "Budapest"]
print(cities.sorted())
```

Array operations

```
let presidents = ["Bush", "Obama", "Trump", "Biden"]
let reversedPresidents = presidents.reversed()
print(reversedPresidents)
```

Dictionaries

```
let employee2 = [
    "name": "Taylor Swift",
    "job": "Singer",
    "location": "Nashville"
]
```

```
print(employee2["name"])
print(employee2["job"])
print(employee2["location"])
```

⚠️ **Expression implicitly coerced from 'String?' To 'Any'**

Dictionaries - safe access

```
print(employee2["name", default: "Unknown"])
print(employee2["job", default: "Unknown"])
print(employee2["location", default: "Unknown"])
```

Creating empty dictionaries

```
var archEnemies = [String: String]()
archEnemies["Batman"] = "The Joker"
archEnemies["Superman"] = "Lex Luthor"
```

Set

```swift
let people = Set([
  "Denzel Washington",
  ”Tom Cruise",
  "Nicolas Cage",
  “Samuel L Jackson”
])

print(people)
```

OR

```swift
var people = Set<String>()
people.insert("Denzel Washington")
people.insert("Tom Cruise")
people.insert("Nicolas Cage")
people.insert("Samuel L Jackson”)
people.insert("Denzel Washington")

print(people)
```

Loops - For

```swift
let platforms = ["iOS", "macOS", "tvOS", "watchOS"]

for os in platforms {
    print("Swift works great on \(os).")
}
```

Loop with range

```swift
for i in 1...5 {
    print("Counting from 1 through 5: \(i)")
}

print()

for i in 1..<5 {
    print("Counting 1 up to 5: \(i)")
}
```

Ignoring loop counter

```
var lyric = "Haters gonna"

for _ in 1...5 {
    lyric += " hate"
}

print(lyric)
```

Loop - While

```swift
var roll = 0

while roll != 20 {
    roll = Int.random(in: 1...20)
    print("I rolled a \(roll)")
}

print("Critical hit!")
```

Continue

```
let filenames = ["me.jpg", "work.txt", "sophie.jpg", "logo.psd"]

for filename in filenames {
    if filename.hasSuffix(".jpg") == false {
        continue
    }

    print("Found picture: \(filename)")
}
```

Break

```swift
let number1 = 4
let number2 = 14
var multiples = [Int]()

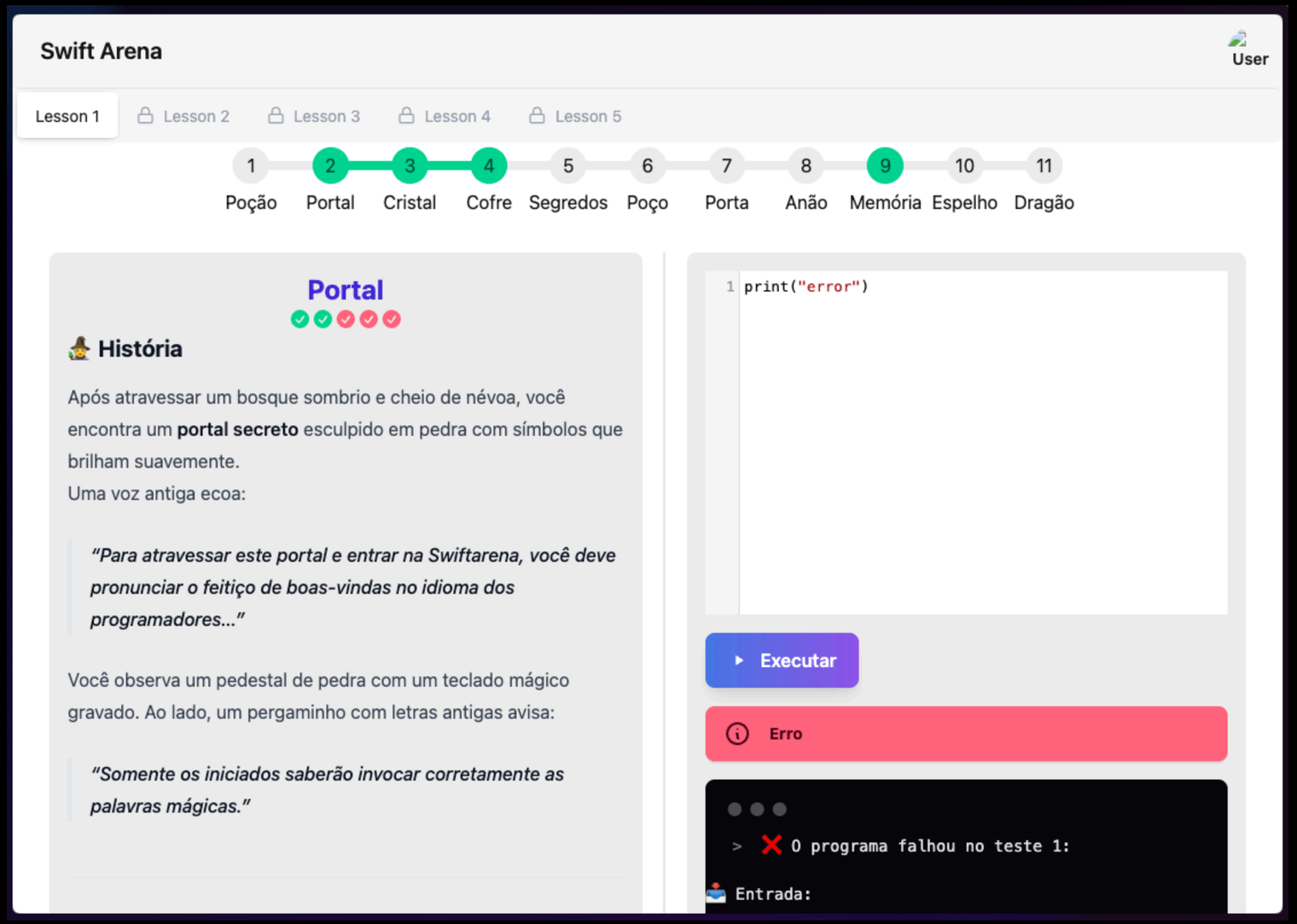
for i in 1...100_000 {
    if i.isMultiple(of: number1) && i.isMultiple(of: number2) {
        multiples.append(i)

        if multiples.count == 10 {
            break
        }
    }
}

print(multiples)
```

Swift Arena

swiftarena.developeracademy.tech



Username: <user@senacsp.edu.br>

Password: <your_senac_id>